Austyn Nguyen

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Experience

Center for Healthcare Engineering & Patient Safety — Ann Arbor, MI

Operations Research Assistant — *Aug. 2025 – Present*

- Partnered with Michigan Medicine's **Chief Transformation Officer** to model inpatient and outpatient patient flow; built cost-impact and capacity models to inform prioritization of a **$50M capital portfolio**.
- Gathered requirements from **3 clinicians** and delivered **20+ automation scripts** to reconfigure **Epic EHR scheduling logic** and improve care access for underserved walk-in patients in a pediatric neurology clinic.

Capital One — McLean, VA

Product Management Intern — *Jun. 2026 – Aug. 2026*

- Collaborated with **4 engineering leads across 2 sprint teams** to launch **2 self-service products** for a **20M+ customer base**; defined UI, data availability, and API requirements and led sprint planning and backlog refinement.
- Produced product requirements documents (PRDs), technical specifications, and objectives and key results (OKRs) for bank statement modernization, targeting $10M in projected savings and $475K in annual operating expense reduction.
- Conducted **28 customer and associate interviews**, call listening, and competitive analysis of **26 services**; mapped **3 customer journeys** and **4 behavioral modes** to prioritize features.
- Coordinated an **associate user acceptance testing (UAT) pilot across 125 test cases** covering navigation, PDF generation, and edge-case account rules; sequenced launch across **3 legacy teams** handling approximately **1,000 manual calls per six months**.

Chemours — Gaithersburg, MD

Data Visualization & Analytics Intern — *May 2025 – Aug. 2025*

- Developed **8 Power BI dashboards** with a **12-member team**, integrating OneLake trade data; translated stakeholder requirements into **DAX models and KPIs** and verified functionality through UAT.
- Optimized equipment reliability reporting across **30+ global manufacturing facilities** by moving transformations upstream to **SQL and Power Query**; cut refresh times by **80%** (5 to 1 minute) and file size by **25%** (400 to 300 MB).
- Established a **Power BI–Git integration framework in DevOps** for a **6-member cross-functional team**; documented workflows and facilitated technical onboarding to standardize version control and dashboard development.

Projects

ML Feedback Categorization Pipeline | *Capital One* — May 2026 – Jul. 2026

- Designed a **two-stage text classification pipeline** combining phrase matching and machine learning to categorize **472 customer comments per month**, saving **6–7 hours of manual review monthly**.
- Evaluated **5 modeling approaches**, including TF-IDF, BERT, large language models (LLMs), and hybrids; increased **precision from 67.2% to 86.4%** and reduced **false positives by 86.5%**; explained tradeoffs to nontechnical stakeholders.

Agentic AI Workflow for Economic Impact Analysis | *Kite & Key Partners* — Jan. 2026 – Apr. 2026

- Architected an **end-to-end agentic AI proof of concept** to research and draft economic impact analyses for **16 concurrent state healthcare bills**, shortening analysis time **from weeks to hours**.
- Co-authored a research paper on **LLM-driven economic analysis** with Institute for Healthcare Policy and Innovation (IHPI) faculty to assess technical feasibility.

Day-Ahead Electricity Price Forecasting Product | *ElectricFish* — Aug. 2025 – Dec. 2025

- Scoped and implemented a **three-stage hybrid machine learning pipeline in Azure ML Studio** to forecast California Independent System Operator (CAISO) day-ahead prices across **10 California cities** for EV charging dynamic pricing.
- Validated on **3 years of CAISO data** using an **80/20 time-series split**; met client acceptance criteria of **R-squared above 0.9, RMSE below 10, and sMAPE below 25** across cities, pricing zones, and peak-load periods.

Education

University of Michigan – Ann Arbor — Ann Arbor, MI

M.S.E. in Industrial and Operations Engineering (Incoming) — *Expected Aug. 2027*

B.S.E. in Computer Science — *Expected Dec. 2026*

- **Notable Coursework:** UI/UX, Conversational AI, Distributed Systems, Databases, Networks

Skills

Product: Product Strategy, Discovery, User Stories, Tech/API Specs, Business Requirements, System Design, Agile/Scrum, Roadmapping, OKRs/KPIs, User Research, Customer Journey, Stakeholder Management, Risk Mitigation, GTM

Technical: Python, SQL, Data Pipelines, Integration, Microsoft Azure, Azure ML Studio, Generative AI, LLMs

Tools: Jira, Figma, Confluence, Power BI (Certified), Claude Code, Codex, Gemini Gems

Campus Involvement: Pops Orchestra (Principal Violist), Chamber Ensemble, Zeta Pi Technical Fraternity